

Project Proposal: Syntactic Resilience and Signal Leakage in Heavily Pruned Transformer Residual Streams

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Research Question:

1)Functional Resilience focuses on the survivor heads to see if remaining Expert heads become more specialized and focused once neighboring heads are removed.To what extent do remaining specialized syntactic heads exhibit increased focus and decreased entropy to maintain performance as redundant heads are pruned?

2)Syntactic Ghosting examines the space between layers to determine if the grammar previously computed by pruned heads leaves a detectable “ESSENCE” signal within the Residual Stream itself.When Expert heads are pruned, is their functional logic compressed into the Residual Stream as a high-frequency signal that later layers can still decode?

3)Functional Taxonomy: Do attention heads naturally cluster into a discrete set of functional types (e.g., local, syntactic, delimiter, or global), and is this behavior controlled by a low-dimensional latent feature space?

4)Behavioral Stability: Are specialized attention behaviors, such as syntactic alignment, statistically stable across diverse sentence architectures, or are they merely artifacts of dataset-level averaging?

Hypothesis:

1) Resilience: Specific Expert Heads act as primary heavy lifters for rules like subject-verb agreement. As the model is pruned from 48 heads down to 5, these heads will show a statistically significant decrease in **Attention Entropy**, becoming laser-focused to compensate for lost neighbors.

2) Linguistic Persistence: Even when Expert Heads are removed, the **Surprisal Gap** will remain low because preceding layers bake grammar rules directly into hidden states. This signal will be detectable via **Linear Probes** and characterized by a drop in the **SVD Participation Ratio** of residual updates.

3) Functional Clustering:Attention heads naturally group into a few recurring categories (Local, Syntactic, Delimiter, and Global). We hypothesize that despite the high head count, variation is controlled by only a few latent dimensions.

4) Behavioral Stability:Specialized attention behaviors (like syntactic alignment) are not merely average trends but are statistically stable across diverse sentence types.

Methodology

To identify where syntactic knowledge resides in a pruned model, this research employs a unified experimental framework using a Sutable LLM Model. The model is systematically reduced using **L_0 Hard Concrete Distributions**, where differentiable gates deactivate non-essential heads across milestones of $48 \rightarrow 30 \rightarrow 20 \rightarrow 10 \rightarrow 5$ active heads.

During this pruning schedule, the following layer measurements are captured:

1. Functional Resilience : Expert heads identified via **Integrated Gradients** for their high attribution to *nsubj* dependencies are monitored for specialization. We track their **Attention Entropy** and **Attention-to-Target** mass, a drop in entropy signals these survivors are becoming laser-focused to compensate for lost neighbors.

2. Syntactic Ghosting: We investigate if pruned logic is compressed into the **Residual Stream**. **Linear Probes** are trained on the frozen update vector ($\Delta h_l = h_l - h_{l-1}$) to predict subject agreement. Simultaneously, we decompose the residual update matrix via **SVD** to calculate the **Participation Ratio**

A sharp drop in Participation Ratio along side high probe accuracy confirms the hypothesis that syntactic signals become lower-dimensional and more concentrated when active heads

3. Functional Taxonomy and Latent Space

To determine if heads cluster into functional types, the following pipeline is executed:

Feature Engineering: For 1,000–5,000 sentences, compute per-head features: offset profiles (-5 to +5), entropy, proportion of attention on previous/next/self-tokens, and syntax-match rates.

Statistical Analysis (R): Standardize the feature matrix and run PCA/SVD to visualize variance explained.

Clustering: Apply k-means or hierarchical clustering to group heads into categories (e.g., local, delimiter, syntactic) and visualize them in PC space to see if layer-depth or syntactic role influences positioning.

4. Stability and Cross-Lingual Evaluation

To ensure specialized behaviors are stable across sentences:

Stability Scoring: For a chosen "Expert" head, compute the fraction of query tokens where the top attention hits the dependency partner across all sentences.

Statistical Estimation: Calculate mean, variance, and bootstrap confidence intervals.

Comparative Analysis: Compare performance across sentence types (short vs. long, simple vs. complex syntax).

Linguistic Contrast: Apply the entire framework to Russian to test if its morphological richness leads to more concentrated ghosts and more distinct functional clusters are removed.

Languages:

English

Russian

5+